

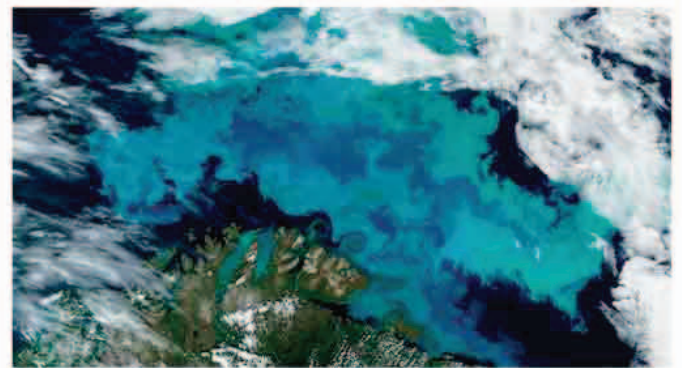


THE BIG FREEZE

Winter air temperatures over the Arctic Ocean sink to below -22°F (-30°C), making the sea freeze over. The ice covers an area up to 5.8 million sq miles (15 million sq km). Most of this melts in summer, leaving less than 2.3 million sq miles (6 million sq km) of ice near the North Pole. Around Antarctica, the winter sea ice covers 8.5 million sq miles (22 million sq km), shrinking in summer to 1.5 million sq miles (4 million sq km).

◀ ICY SEAS

As the spring Sun warms the sea near Baffin Island in Arctic Canada, the sea ice starts breaking up and drifting along the coast with the swirling currents. By midsummer (inset), all the sea ice has melted.



PLANKTON BLOOM

The water of cold oceans is rich in minerals stirred up from near the seabed. These are vital nutrients for the tiny algae of the phytoplankton. When the ice melts in summer, they combine with the 24-hour daylight to fuel huge blooms of phytoplankton, as seen here (blue) in the Arctic. In turn, these provide food for other marine life.



EBB AND FLOW

As the polar sea ice forms and then melts away with the seasons, its edge is constantly moving north and south. The mobile ice front is a food-rich zone that attracts many polar animals, including these beluga whales swimming through a gap in the floating ice to reach nearby open water.

